AERIS Expert Review Packet

Anomaly

Source / metric	tceq / so2
Observed / expected	13.5 / -0.393827
Time	2026-06-21 13:00 UTC
Location	29.7337, -95.2576
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 606	52.47 to 100; mean 84.5674	100 at 2026-06-21 12:55Z; dt 5 min
asos	temperature	degC	9 / 606	23 to 34.4444; mean 27.736	24 at 2026-06-21 12:55Z; dt 5 min
asos	wind_direction	degree	9 / 607	0 to 270; mean 132.175	200 at 2026-06-21 12:55Z; dt 5 min
asos	wind_speed	m/s	9 / 607	0 to 10.8033; mean 3.4867	5.14444 at 2026-06-21 12:55Z; dt 5 min
noaa_gfs	gh_500	m	10 / 120	5875.31 to 5946.6; mean 5900.79	5875.51 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	pbl_height	m	10 / 120	49.5963 to 1968.25; mean 642.553	323.367 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	10 / 120	36.8269 to 60.3207; mean 50.3339	58.3479 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	10 / 120	1004.61 to 1015.32; mean 1009.74	1009.12 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	t_850	degC	10 / 120	17.2542 to 22.3564; mean 19.4632	18.3753 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	u_10m	m/s	10 / 120	-3.27924 to 1.55084; mean -0.9026	0.370845 at 2026-06-21 12:00Z; dt 60 min
noaa_gfs	v_10m	m/s	10 / 120	-1.84918 to 6.63198; mean 3.3653	3.21198 at 2026-06-21 12:00Z; dt 60 min
openaq	ozone	ppm	17 / 1022	-0.002 to 0.036; mean 0.0148	0.013 at 2026-06-21 13:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 217	6 to 95; mean 29.7512	16 at 2026-06-21 13:00Z; dt 0 min
openaq	pm25	ug/m3	82 / 2920	0 to 93.73; mean 6.0735	8 at 2026-06-21 13:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 56.7056	52 at 2026-06-21 13:00Z; dt 1 min
openweather	humidity	percent	5 / 360	58 to 97; mean 85.2944	96 at 2026-06-21 13:00Z; dt 1 min
openweather	precipitation	mm/h	5 / 360	0 to 4.74; mean 0.3508	1.69 at 2026-06-21 13:00Z; dt 1 min
openweather	pressure	hPa	5 / 360	1009 to 1016; mean 1012.05	1011 at 2026-06-21 13:00Z; dt 1 min
openweather	temperature	degC	5 / 360	23.58 to 34.66; mean 28.1414	24.34 at 2026-06-21 13:00Z; dt 1 min
openweather	wind_direction	degree	5 / 360	0 to 351; mean 157.328	0 at 2026-06-21 13:00Z; dt 1 min
openweather	wind_speed	m/s	5 / 360	0.45 to 6.56; mean 2.4729	1.79 at 2026-06-21 13:00Z; dt 1 min
purpleair	pm25	ug/m3	67 / 4714	0 to 359.8; mean 7.612	5.6 at 2026-06-21 13:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-21 18:26Z; dt 326.9 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-21 18:26Z; dt 326.9 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.0247994 to 0.0279886; mean 0.0264	0.0279886 at 2026-06-21 18:26Z; dt 326.9 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-21 18:26Z; dt 326.9 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	9.69566e-05 to 0.000168978; mean 0.0001	0.000168978 at 2026-06-21 18:56Z; dt 356.4 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-21 18:56Z; dt 356.4 min
sentinel5p	s5p_no2_column	mol/m^2	3 / 3	1.683e-05 to 3.59018e-05; mean 0	3.59018e-05 at 2026-06-21 18:56Z; dt 356.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-06-21 18:56Z; dt 356.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-21 18:26Z; dt 326.9 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-0.00020099 to 1.36867e-05; mean -0	-4.39631e-06 at 2026-06-21 18:56Z; dt 356.4 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-21 18:56Z; dt 356.4 min
tceq	co	ppm	3 / 212	0 to 0.8; mean 0.2458	0.2 at 2026-06-21 13:00Z; dt 0 min
tceq	no2	ppb	12 / 818	-2.4 to 24.5; mean 4.2511	0.6 at 2026-06-21 13:00Z; dt 0 min
tceq	so2	ppb	3 / 207	-1.6 to 13.5; mean -0.2237	13.5 at 2026-06-21 13:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-20 00Z 88.08, 06Z 92.31, 12Z 84.14, 18Z 91.64, 06-21 00Z 93.71, 06Z 96.51, 12Z 88.39, 18Z 71.07, 06-22 00Z 85.81, 06Z 91.7, 12Z 73.85, 18Z 61.24, 06-23 00Z 74.45
asos	temperature	06-20 00Z 27.41, 06Z 26.82, 12Z 28.28, 18Z 24.66, 06-21 00Z 24.52, 06Z 24.93, 12Z 25.89, 18Z 30.87, 06-22 00Z 27.96, 06Z 27.3, 12Z 30.71, 18Z 32.6, 06-23 00Z 29.72
asos	wind_direction	06-20 00Z 129.2, 06Z 119, 12Z 60.21, 18Z 124, 06-21 00Z 50.83, 06Z 84.2, 12Z 161.7, 18Z 166.6, 06-22 00Z 155.4, 06Z 164.8, 12Z 174.4, 18Z 171.2, 06-23 00Z 168.8
asos	wind_speed	06-20 00Z 3.653, 06Z 2.486, 12Z 2.111, 18Z 2.851, 06-21 00Z 1.093, 06Z 1.883, 12Z 3.789, 18Z 5.504, 06-22 00Z 3.925, 06Z 3.23, 12Z 4.925, 18Z 5.659, 06-23 00Z 5.273
noaa_gfs	gh_500	06-20 06Z 5903, 12Z 5881, 18Z 5896, 06-21 00Z 5882, 06Z 5885, 12Z 5876, 18Z 5890, 06-22 00Z 5894, 06Z 5915, 12Z 5908, 18Z 5934, 06-23 00Z 5946
noaa_gfs	pbl_height	06-20 06Z 530.1, 12Z 240.9, 18Z 927.8, 06-21 00Z 186.6, 06Z 140.2, 12Z 299, 18Z 616.7, 06-22 00Z 879, 06Z 552.2, 12Z 547, 18Z 1745, 06-23 00Z 1046
noaa_gfs	precipitable_water	06-20 06Z 55.09, 12Z 52.71, 18Z 55.49, 06-21 00Z 53.48, 06Z 56.2, 12Z 57.95, 18Z 56.17, 06-22 00Z 53.65, 06Z 43.63, 12Z 42.18, 18Z 39.6, 06-23 00Z 37.86
noaa_gfs	surface_pressure	06-20 06Z 1010, 12Z 1010, 18Z 1011, 06-21 00Z 1008, 06Z 1009, 12Z 1008, 18Z 1009, 06-22 00Z 1007, 06Z 1009, 12Z 1010, 18Z 1013, 06-23 00Z 1012
noaa_gfs	t_850	06-20 06Z 19.91, 12Z 18.56, 18Z 18.73, 06-21 00Z 21.45, 06Z 19.48, 12Z 18.62, 18Z 19.51, 06-22 00Z 20.18, 06Z 19.92, 12Z 18.85, 18Z 17.85, 06-23 00Z 20.48
noaa_gfs	u_10m	06-20 06Z -1.474, 12Z -0.782, 18Z -2.906, 06-21 00Z -1.932, 06Z -0.783, 12Z 0.4488, 18Z -0.4373, 06-22 00Z -0.9744, 06Z -0.617, 12Z -0.4874, 18Z -0.0208, 06-23 00Z -0.8663

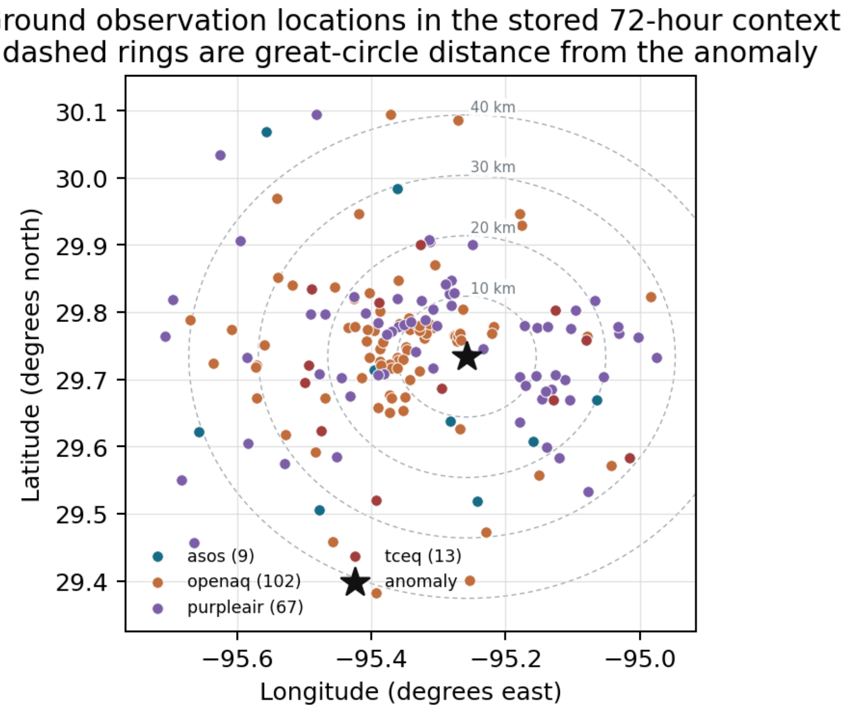
Source	Metric	Values
noaa_gfs	v_10m	06-20 06Z 4.174, 12Z 2.492, 18Z 1.042, 06-21 00Z 0.04582, 06Z 1.889, 12Z 3.182, 18Z 3.395, 06-22 00Z 5.284, 06Z 4.083, 12Z 4.048, 18Z 5.175, 06-23 00Z 5.574
openaq	ozone	06-20 00Z 0.01984, 06Z 0.01288, 12Z 0.01231, 18Z 0.0135, 06-21 00Z 0.008507, 06Z 0.003192, 12Z 0.02052, 18Z 0.0218, 06-22 00Z 0.01395, 06Z 0.01341, 12Z 0.01596, 18Z 0.01922, 06-23 00Z 0.01744
openaq	pm10	06-20 00Z 21.2, 06Z 30.39, 12Z 30.44, 18Z 11.61, 06-21 00Z 19.28, 06Z 31.67, 12Z 11.19, 18Z 21.61, 06-22 00Z 34.44, 06Z 36.06, 12Z 55, 18Z 49.61, 06-23 00Z 32.83
openaq	pm25	06-20 00Z 6.617, 06Z 5.859, 12Z 5.44, 18Z 3.118, 06-21 00Z 4.108, 06Z 6.943, 12Z 2.632, 18Z 4.409, 06-22 00Z 5.88, 06Z 10.04, 12Z 13.92, 18Z 4.518, 06-23 00Z 3.511
openweather	cloud_cover	06-20 00Z 99.08, 06Z 94.23, 12Z 99.5, 18Z 99.61, 06-21 00Z 98.57, 06Z 41.63, 12Z 47.2, 18Z 75.93, 06-22 00Z 10.37, 06Z 7.2, 12Z 17.83, 18Z 5.31, 06-23 00Z 3.333
openweather	humidity	06-20 00Z 88, 06Z 91.13, 12Z 86.37, 18Z 82.23, 06-21 00Z 93.8, 06Z 95.77, 12Z 90.87, 18Z 74.83, 06-22 00Z 84.23, 06Z 91.17, 12Z 80.9, 18Z 66.86, 06-23 00Z 72.33
openweather	precipitation	06-20 00Z 0.1329, 06Z 0.43, 12Z 1.097, 18Z 0.7358, 06-21 00Z 0.01667, 06Z 0.1767, 12Z 1.158, 18Z 0.464, 06-22 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-23 00Z 0
openweather	pressure	06-20 00Z 1012, 06Z 1012, 12Z 1014, 18Z 1012, 06-21 00Z 1011, 06Z 1011, 12Z 1011, 18Z 1010, 06-22 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1014, 06-23 00Z 1014
openweather	temperature	06-20 00Z 27.51, 06Z 27.41, 12Z 28.55, 18Z 27.55, 06-21 00Z 24.3, 06Z 24.69, 12Z 25.49, 18Z 31.42, 06-22 00Z 28.69, 06Z 27.46, 12Z 30.33, 18Z 33.66, 06-23 00Z 31.61
openweather	wind_direction	06-20 00Z 132.3, 06Z 164.8, 12Z 115.8, 18Z 161.4, 06-21 00Z 121.6, 06Z 168.9, 12Z 165.7, 18Z 175.7, 06-22 00Z 157.9, 06Z 163.4, 12Z 177.4, 18Z 174.3, 06-23 00Z 178
openweather	wind_speed	06-20 00Z 1.595, 06Z 2.456, 12Z 1.399, 18Z 3.311, 06-21 00Z 1.668, 06Z 1.597, 12Z 3.18, 18Z 3.07, 06-22 00Z 2.593, 06Z 2.518, 12Z 3.077, 18Z 2.897, 06-23 00Z 3.095
purpleair	pm25	06-20 00Z 7.661, 06Z 6.934, 12Z 6.522, 18Z 4.924, 06-21 00Z 5.871, 06Z 8.427, 12Z 5.627, 18Z 5.958, 06-22 00Z 7.425, 06Z 13.79, 12Z 13.15, 18Z 5.284, 06-23 00Z 6.163
sentinel5p	s5p_aer_ai_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_ch4_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_co_column	06-21 18Z 0.0279, 06-22 18Z 0.02489
sentinel5p	s5p_co_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_hcho_column	06-21 18Z 0.000131, 06-22 18Z 9.933e-05
sentinel5p	s5p_hcho_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_no2_column	06-21 18Z 2.637e-05, 06-22 18Z 2.646e-05
sentinel5p	s5p_no2_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_o3_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
sentinel5p	s5p_so2_column	06-21 18Z -8.157e-05, 06-22 18Z 5.768e-06
sentinel5p	s5p_so2_granule_available	06-20 18Z 1, 06-21 18Z 1, 06-22 18Z 1
tceq	co	06-20 00Z 0.3, 06Z 0.1667, 12Z 0.2, 18Z 0.2944, 06-21 00Z 0.3556, 06Z 0.2611, 12Z 0.2059, 18Z 0.2, 06-22 00Z 0.2222, 06Z 0.2278, 12Z 0.2778, 18Z 0.2333, 06-23 00Z 0.2333
tceq	no2	06-20 00Z 7.805, 06Z 2.27, 12Z 4.406, 18Z 6.483, 06-21 00Z 10.06, 06Z 5.241, 12Z 4.138, 18Z 1.519, 06-22 00Z 2.099, 06Z 2.53, 12Z 3.778, 18Z 2.265, 06-23 00Z 2.263
tceq	so2	06-20 00Z 0.02667, 06Z -0.1, 12Z -0.06111, 18Z -0.1, 06-21 00Z -0.1273, 06Z -0.5, 12Z 0.4294, 18Z -0.4222, 06-22 00Z -0.4, 06Z -0.4556, 12Z -0.2857, 18Z -0.4722, 06-23 00Z -0.5667

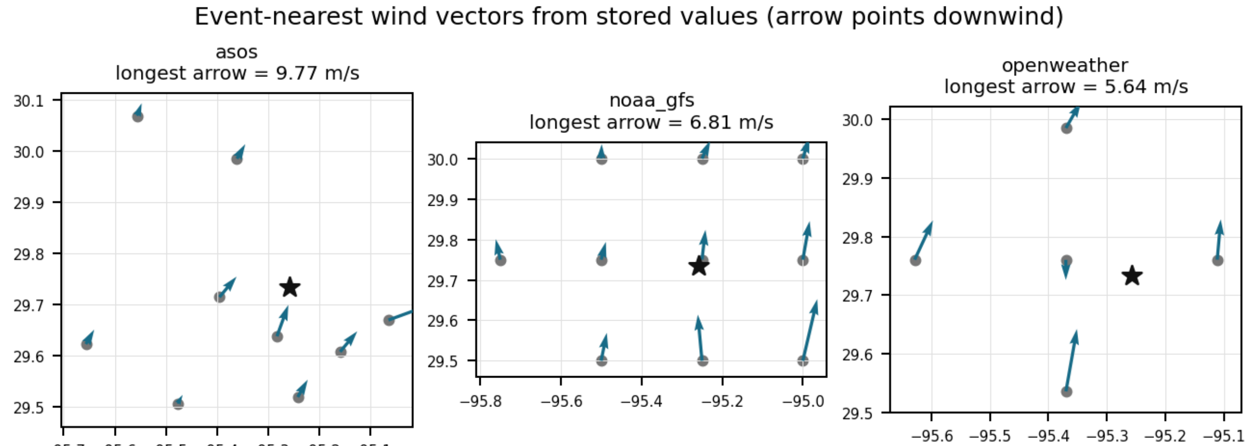
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	HOU (10.965 km) 84.05, MCJ (13.448 km) 86.39, EFD (16.998 km) 84.05, T41 (20.02 km) 84.63, LVJ (23.937 km) 84.19, IAH (29.593 km) 85.71, AXH (33.015 km) 80.98, SGR (40.48 km) 85.86, +1 more
asos	temperature	HOU (10.965 km) 27.89, MCJ (13.448 km) 27.56, EFD (16.998 km) 28.5, T41 (20.02 km) 27.94, LVJ (23.937 km) 27.73, IAH (29.593 km) 27.54, AXH (33.015 km) 27.78, SGR (40.48 km) 27.76, +1 more
asos	wind_direction	HOU (10.965 km) 141.4, MCJ (13.448 km) 151.5, EFD (16.998 km) 157.8, T41 (20.02 km) 153.2, LVJ (23.937 km) 118.2, IAH (29.593 km) 119, AXH (33.015 km) 110.4, SGR (40.48 km) 135.4, +1 more
asos	wind_speed	HOU (10.965 km) 4.008, MCJ (13.448 km) 5.166, EFD (16.998 km) 4.968, T41 (20.02 km) 4.123, LVJ (23.937 km) 2.751, IAH (29.593 km) 3.272, AXH (33.015 km) 1.936, SGR (40.48 km) 3.78, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.25 (1.952 km) 5901, gfs:29.75,-95.50 (23.473 km) 5901, gfs:29.75,-95.00 (24.937 km) 5901, gfs:29.50,-95.25 (26.0 km) 5902, gfs:30.00,-95.25 (29.617 km) 5900, gfs:29.50,-95.50 (34.993 km) 5901, gfs:29.50,-95.00 (35.994 km) 5902, gfs:30.00,-95.50 (37.722 km) 5900, +2 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (1.952 km) 675.1, gfs:29.75,-95.50 (23.473 km) 700.1, gfs:29.75,-95.00 (24.937 km) 635.2, gfs:29.50,-95.25 (26.0 km) 580.6, gfs:30.00,-95.25 (29.617 km) 697.9, gfs:29.50,-95.50 (34.993 km) 580.3, gfs:29.50,-95.00 (35.994 km) 658.6, gfs:30.00,-95.50 (37.722 km) 751, +2 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (1.952 km) 50.51, gfs:29.75,-95.50 (23.473 km) 50.11, gfs:29.75,-95.00 (24.937 km) 50.88, gfs:29.50,-95.25 (26.0 km) 50.53, gfs:30.00,-95.25 (29.617 km) 50.26, gfs:29.50,-95.50 (34.993 km) 50.14, gfs:29.50,-95.00 (35.994 km) 50.82, gfs:30.00,-95.50 (37.722 km) 49.73, +2 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (1.952 km) 1010, gfs:29.75,-95.50 (23.473 km) 1009, gfs:29.75,-95.00 (24.937 km) 1011, gfs:29.50,-95.25 (26.0 km) 1011, gfs:30.00,-95.25 (29.617 km) 1009, gfs:29.50,-95.50 (34.993 km) 1010, gfs:29.50,-95.00 (35.994 km) 1012, gfs:30.00,-95.50 (37.722 km) 1007, +2 more
noaa_gfs	t_850	gfs:29.75,-95.25 (1.952 km) 19.37, gfs:29.75,-95.50 (23.473 km) 19.47, gfs:29.75,-95.00 (24.937 km) 19.56, gfs:29.50,-95.25 (26.0 km) 19.49, gfs:30.00,-95.25 (29.617 km) 19.31, gfs:29.50,-95.50 (34.993 km) 19.6, gfs:29.50,-95.00 (35.994 km) 19.61, gfs:30.00,-95.50 (37.722 km) 19.34, +2 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (1.952 km) -0.8638, gfs:29.75,-95.50 (23.473 km) -0.8138, gfs:29.75,-95.00 (24.937 km) -0.9321, gfs:29.50,-95.25 (26.0 km) -1.038, gfs:30.00,-95.25 (29.617 km) -0.8738, gfs:29.50,-95.50 (34.993 km) -1.039, gfs:29.50,-95.00 (35.994 km) -0.8371, gfs:30.00,-95.50 (37.722 km) -0.628, +2 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (1.952 km) 2.984, gfs:29.75,-95.50 (23.473 km) 3.055, gfs:29.75,-95.00 (24.937 km) 3.273, gfs:29.50,-95.25 (26.0 km) 3.855, gfs:30.00,-95.25 (29.617 km) 2.876, gfs:29.50,-95.50 (34.993 km) 3.409, gfs:29.50,-95.00 (35.994 km) 4.592, gfs:30.00,-95.50 (37.722 km) 3.123, +2 more
openaq	ozone	3378 (0.021 km) 0.01155, 2039 (5.231 km) 0.01304, 325 (6.369 km) 0.01886, 2046 (10.803 km) 0.0158, 320 (12.059 km) 0.01183, 1319333 (14.033 km) 0.0167, 332 (14.34 km) 0.01881, 3214 (14.86 km) 0.01528, +9 more
openaq	pm10	13380082 (0.021 km) 31.83, 15852419 (7.017 km) 34.11, 271 (14.34 km) 23.37
openaq	pm25	15539682 (0.005 km) 7.745, 3379 (0.021 km) 9.483, 8967123 (0.058 km) 9.428, 8967479 (0.058 km) 8.046, 9489522 (2.89 km) 6.374, 8967065 (2.933 km) 7.253, 9201801 (3.625 km) 4.841, 12178865 (3.713 km) 7.378, +74 more
openweather	cloud_cover	grid:center (11.237 km) 55.53, grid:east (14.451 km) 56.71, grid:south (24.535 km) 60.17, grid:north (29.962 km) 55.29, grid:west (35.93 km) 55.83
openweather	humidity	grid:center (11.237 km) 84.64, grid:east (14.451 km) 87.07, grid:south (24.535 km) 84.68, grid:north (29.962 km) 85.47, grid:west (35.93 km) 84.61
openweather	precipitation	grid:center (11.237 km) 0.2703, grid:east (14.451 km) 0.3461, grid:south (24.535 km) 0.3368, grid:north (29.962 km) 0.4086, grid:west (35.93 km) 0.3919
openweather	pressure	grid:center (11.237 km) 1012, grid:east (14.451 km) 1012, grid:south (24.535 km) 1012, grid:north (29.962 km) 1012, grid:west (35.93 km) 1012
openweather	temperature	grid:center (11.237 km) 28.32, grid:east (14.451 km) 28.11, grid:south (24.535 km) 28.27, grid:north (29.962 km) 27.94, grid:west (35.93 km) 28.08
openweather	wind_direction	grid:center (11.237 km) 143, grid:east (14.451 km) 166.9, grid:south (24.535 km) 179.6, grid:north (29.962 km) 136.9, grid:west (35.93 km) 160.2
openweather	wind_speed	grid:center (11.237 km) 2.299, grid:east (14.451 km) 3.612, grid:south (24.535 km) 2.196, grid:north (29.962 km) 2.216, grid:west (35.93 km) 2.041

Source	Metric	Values
purpleair	pm25	99797 (0.6 km) 6.543, 166435 (2.677 km) 9.997, 6752 (5.276 km) 7.563, 222601 (6.699 km) 4.788, 133994 (7.377 km) 0.6575, 235425 (8.237 km) 7.056, 229853 (8.54 km) 5.249, 127441 (8.731 km) 9.448, +59 more
tceq	co	482011035 (0.0 km) 0.1557, 482011039 (14.355 km) 0.1676, 482011052 (15.44 km) 0.4127
tceq	no2	482011035 (0.0 km) 5.966, 482010416 (6.378 km) 2.575, 482011039 (14.355 km) 3.756, 482010026 (14.878 km) 6.634, 482011052 (15.44 km) 9.245, 482011015 (17.437 km) 4.317, 482010024 (19.743 km) 1.161, 482011066 (22.737 km) 4.804, +4 more
tceq	so2	482011035 (0.0 km) -0.09565, 482011039 (14.355 km) 0.1714, 482010051 (24.236 km) -0.7603





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

The anomalous sulfur dioxide value occurred at the same monitor and hour when nearby tceq co was 0.2 ppm and tceq no2 was 0.6 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

The CO, NO2, and SO2 levels measured by Sentinel-5P and TCEQ monitors were within normal ranges, indicating that the anomaly was not caused by significant emissions or atmospheric changes.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

A broad regional sulfur pollution episode is contradicted by the available regional indicators because satellite SO2 later on 2026-06-21 did not show a positive column enhancement over the area and particle concentrations near the anomaly time were low to moderate rather than regionally elevated across OpenAQ and PurpleAir sensors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The TCEQ monitoring network recorded a severe sulfur dioxide (SO2) concentration spike reaching 13.5 ppb at coordinates (29.734, -95.258) on June 21, 2026, at 13:00:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

ASOS surface observations around 13:00 UTC on June 21, 2026, showed high relative humidity and southerly to southeasterly winds near the anomaly site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Meteorological trapping was likely a contributing amplifier but not the primary cause because humidity and precipitable water were high and the boundary layer was still relatively shallow near the event time, yet surface winds were not calm and no coincident regional increases appeared in particulate matter or ozone measurements.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

A shallow morning planetary boundary layer compressed atmospheric volume near the ground, enhancing surface SO_2 accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

The anomaly occurred on June 21, 2026, with the highest PM2.5 levels recorded between 06Z and 12Z.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Local wind flow from the south and southeast transported industrial sulfur emissions toward the monitoring site during the morning hours.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

A refinery or petrochemical upset event such as flaring, sulfur recovery unit release, or stack plume interception is a plausible physical cause because the anomaly was extremely localized to one TCEQ sulfur dioxide monitor and occurred under southerly to south-southwesterly low-to-moderate winds that could advect emissions from the Houston Ship Channel industrial corridor toward the site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The severe SO2 anomaly reaching 13.5 ppb recorded by TCEQ was localized to a single monitoring station, as regional satellite total column SO2 measurements from Sentinel-5P and surrounding TCEQ ground monitors maintained near-zero baseline levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

A severe thunderstorm or tornado event has stirred up pollutants and particles from the ground, contributing to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

Point-source emissions from industrial and petrochemical facilities along the Houston Ship Channel likely caused the elevated SO2 spike recorded by TCEQ monitors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

NOAA GFS model data indicated a shallow planetary boundary layer height of approximately 323 meters near the anomaly location at 12:00 UTC on June 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The temperature and wind speed data from OpenWeather suggest that the anomaly occurred during a period of relatively calm weather.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

The air quality anomaly in Houston, Texas was characterized by elevated PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The pollutant with the detected anomaly is sulfur dioxide measured by tceq.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

Meteorological transport and dispersion conditions are consistent with a local plume reaching the monitor because surface winds near the event were southerly to south-southwesterly at roughly 1.8 to 5.1 m/s and the boundary-layer height was only about 323 m near 12Z, which would favor advection from nearby sources with only modest vertical dilution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Meteorology and remote sensing are most consistent with a short-duration local industrial SO_2 plume because winds near the event were southerly to south-southwesterly with moderate speeds, gfs showed a relatively shallow boundary layer near 299 to 323 m around 12Z, and sentinel5p later showed no broad SO_2 column enhancement over the area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

ASOS surface meteorology recorded high relative humidity near 100 percent and southerly to southeasterly winds averaging 3.8 to 5.5 m/s at the time of the SO_2 spike, supporting potential local transport from nearby industrial sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

NOAA GFS model output indicated a low planetary boundary layer height of approximately 323 meters near the anomaly location at 12:00:00 UTC on June 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

A short-duration sulfur dioxide emission from a nearby industrial combustion or process unit is the most plausible physical cause because the TCEQ sulfur dioxide concentration reached 13.5 ppb at the anomaly monitor while the colocated TCEQ carbon monoxide and nitrogen dioxide remained low, which is consistent with a sulfur-rich plume rather than a broad combustion pollution episode.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

Surface meteorological observations from ASOS near the anomaly location at 12:55:00 UTC on June 21, 2026, indicated south-southeasterly winds of approximately 5.1 m/s and relative humidity reaching 100 percent.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and accumulation of pollutants.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

TCEQ monitor 482011035 in Houston recorded an SO_2 concentration peak of 13.5 ppb at 13:00 UTC on June 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Local co-pollutant measurements do not support a broad combustion or regional haze episode because the collocated TCEQ monitor reported CO of 0.2 ppm and NO_2 of 0.6 ppb at 2026-06-21T13:00:00+00:00 while nearby openaq and purpleair particulate measurements remained modest.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

A short-duration local sulfur-rich industrial plume is strongly supported because the TCEQ SO_2 monitor at the anomaly site measured 13.5 ppb at 2026-06-21T13:00:00+00:00 while the same monitor's expected value was near zero and nearby TCEQ CO and NO_2 at the same site remained low at 0.2 ppm and 0.6 ppb, which matches an isolated SO_2 enhancement rather than a general combustion plume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

The TCEQ monitor at 29.734, -95.258 recorded a sharply isolated SO_2 spike to 13.5 ppb at 2026-06-21T13:00:00+00:00 while the 6-hour mean SO_2 across TCEQ monitors for 06-21 12Z was only 0.4294 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

GFS atmospheric model data showed a reduced planetary boundary layer height near 300 meters during the morning of June 21, 2026, which restricted vertical atmospheric mixing and trapped localized SO_2 emissions near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The PM2.5 levels are elevated, with a mean of 7.612 ug/m3 and a nearest value of 5.6 ug/m3 at 2026-06-21T13:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

The tceq sulfur dioxide measurement at 2026-06-21T13:00:00+00:00 at 29.734, -95.258 was 13.5 ppb, compared with an expected value of -0.39382716049382716 ppb, and the reported z-score was 145.66530461416133 with severity classified as severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.